

HALLIBURTON FIELD, OK, USA

WMO: 722078

Lat: 34.471N

Lon: 97.960W

Elev: 1113

StdP: 14.11

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53938

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.7	21.0	5.0	7.4	22.0	10.1	9.6	27.3	27.1	43.9	25.2	45.1	8.3	10	0.503

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.8	103.4	74.4	99.3	74.7	96.7	74.6	78.7	90.4	77.8	90.4	77.0	89.7	10.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.4	138.7	82.7	74.7	135.2	82.0	73.2	128.7	81.0	43.0	91.2	42.0	89.7	41.1	90.1	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.7	21.5	19.3	9.5	104.9	4.5	3.9	6.2	107.7	3.5	110.0	1.0	112.1	-2.3	114.9
			8.3	80.5	4.2	1.6	5.3	81.7	2.9	82.6	0.5	83.5	-2.5	84.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.9	41.5	44.8	55.0	62.2	70.1	79.1	83.5	83.3	75.4	63.7	52.8	42.2
	DBStd	17.04	9.48	10.90	10.05	8.23	7.64	4.30	5.04	5.73	6.19	8.51	9.04	9.50
	HDD50	929	290	208	65	11	1	0	0	0	0	7	73	274
	HDD65	3026	728	567	333	145	40	0	0	0	5	128	373	707
	CDD50	5637	27	63	218	376	624	874	1039	1031	762	433	158	32
	CDD65	2260	0	3	21	59	198	424	574	566	317	89	8	1
	CDH74	25597	6	35	205	594	1841	4573	7309	7043	3095	805	85	6
	CDH80	12878	0	5	38	155	676	2165	4176	4024	1398	231	10	0
Wind	WSAvg	8.7	8.4	9.3	10.6	11.0	9.6	8.9	7.4	7.6	7.4	7.9	8.5	8.4
Precipitation	PrecAvg	35.4	1.5	1.8	2.8	3.6	4.7	4.2	2.5	2.9	3.3	3.5	2.4	2.0
	PrecMax	46.7	3.6	5.2	6.7	9.0	10.4	13.0	6.7	7.4	8.2	11.7	8.0	6.5
	PrecMin	23.1	0.0	0.0	0.1	0.4	0.6	1.1	0.1	0.0	0.1	0.2	0.1	0.1
	PrecStd	6.9	1.0	1.4	1.9	2.1	2.6	2.6	2.0	1.9	2.2	2.6	2.2	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.4	79.1	84.2	90.0	95.1	98.9	107.8	108.7	100.0	90.3	81.4	72.9
		MCWB	55.0	59.5	64.8	66.5	72.1	75.2	73.8	74.4	71.9	71.5	64.8	57.4
	2%	DB	68.1	72.6	79.3	84.0	90.5	95.3	103.7	104.1	93.4	85.9	75.8	66.5
		MCWB	53.5	57.5	62.6	65.4	73.0	75.0	74.5	74.4	73.2	69.5	61.1	55.3
	5%	DB	63.4	67.7	75.1	80.9	87.8	92.8	99.3	99.8	90.7	81.8	72.2	63.0
		MCWB	52.2	56.4	60.7	65.0	72.0	74.9	74.8	74.9	72.9	67.3	60.0	53.6
	10%	DB	58.5	63.0	71.8	77.0	83.0	90.6	96.8	96.9	88.3	78.9	67.9	57.3
		MCWB	48.1	53.5	59.4	63.6	70.9	74.7	75.1	74.7	72.3	65.5	57.8	49.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.4	64.5	69.4	72.8	77.5	80.2	80.1	80.0	78.2	75.3	68.7	63.8
		MCDB	66.6	70.8	75.8	82.3	88.7	89.9	91.1	91.0	89.0	82.6	75.8	68.4
	2%	WB	57.8	61.4	66.8	70.6	75.6	78.2	78.4	78.4	76.5	73.4	65.1	60.0
		MCDB	64.0	67.0	75.1	78.4	85.0	89.0	90.9	91.0	86.9	81.4	71.4	63.7
	5%	WB	52.7	58.5	64.4	68.7	74.3	77.0	77.6	77.4	75.2	71.5	62.8	54.7
		MCDB	61.0	65.4	70.6	75.6	82.8	88.2	91.0	91.2	84.9	78.1	67.4	59.4
	10%	WB	48.6	54.0	62.1	66.7	72.8	75.8	76.9	76.4	74.1	69.2	60.2	50.7
		MCDB	58.0	62.0	68.2	73.7	80.7	86.6	90.3	90.5	83.6	74.9	66.4	57.0
Mean Daily Temperature Range	5% DB	MDBR	22.8	22.0	22.6	22.9	20.2	20.2	21.8	21.8	21.3	22.4	22.9	20.6
		MCDBR	31.3	29.8	27.2	27.8	24.3	23.2	26.1	26.4	24.6	25.5	27.2	28.0
		MCWBR	19.1	17.2	14.3	14.3	10.2	7.6	6.4	6.5	8.8	12.2	15.3	17.7
	5% WB	MCDBR	27.5	26.5	21.7	21.5	19.9	20.1	21.0	22.0	19.7	19.4	22.0	21.8
MCWBR		18.9	18.0	12.2	13.3	10.3	8.2	7.2	7.1	8.6	11.0	15.5	17.1	
Clear-Sky Solar Irradiance	taub	0.293	0.301	0.339	0.372	0.408	0.426	0.423	0.419	0.386	0.348	0.315	0.295	
	taud	2.514	2.507	2.418	2.352	2.319	2.309	2.351	2.308	2.375	2.470	2.495	2.508	
	Ebn at Noon	293	302	297	290	279	272	272	272	277	281	281	285	
	Edh at Noon	26	29	35	39	41	41	39	40	36	30	26	25	
All-Sky Solar Radiation	RadAvg	921	1105	1463	1794	1917	2149	2174	1987	1649	1267	973	785	
	RadStd	95	162	93	144	203	196	128	114	146	179	123	89	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page